



2024

X19

BOOKLET NO.

160026

Forest Services
Mechanical Engineering

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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SECTION - A

Q1. Solve **any five** out of seven.

(8×5=40)

- (a) An overhang crank with pin and shaft is shown in Fig. A. Tangential load of 15 kN acts on the crank pin. Determine the maximum stress (principal) and maximum shear stress at the centre of the crank shaft bearing.

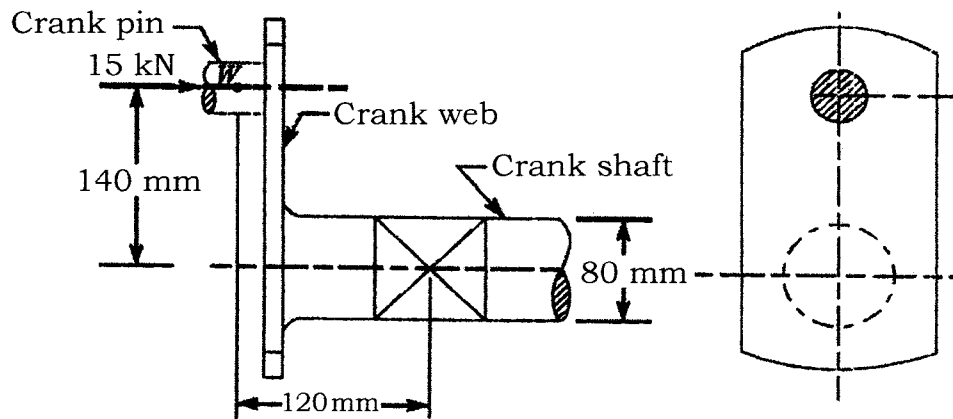


Fig. A

- (b) Discuss the importance of crystal structure and its reflections on properties of materials.
- (c) Classify metal forming processes and differentiate between hot working and cold working.
- (d) Discuss the purpose of sensor in mechatronics and its different characteristics.
- (e) Explain the purpose of PERT and CPM in project management and key components of a network diagram.
- (f) Imagine you are a Production Manager at a factory that produces two types of machines – Machine A and Machine B. The factory has certain resource constraints (such as labor hours and materials) that limit the number of machines that can be produced. Company wish to determine the units of Machine A and Machine B to be produce to improve the utilization of resources (labor hours and materials).

Machine A requires :

- 5 hrs of labor
- 10 units of raw material



Machine B requires :

- 3 hrs of labor
- 8 units of raw material

The factory has :

- A total of 150 hours of labor available.
- A total of 120 units of raw material available.

Answer the following :

- i) Define linear system of equations with suitable variables.
 - ii) Write a pseudocode in 'C' to find solution of system of linear equations.
 - iii) Give sample calculation to test the feasibility of solution.
 - iv) Give significance of unique solution in system of linear equation.
- (g) Consider the integral of a simple function, say $f(x) = x^2$ over the interval $[0, 1]$ using the trapezoidal rule.

The exact solution can be found as $\int_0^1 x^2 = \frac{1}{3}$.

Answer the following :

- i) What is the significance of uniform portion in numerical integration ?
- ii) Write a pseudocode in 'C' to approximate the given integral using trapezoidal rule.
- iii) In numerical integration, why does increasing the number of sub intervals improve the accuracy ?

Q2. (a) A pair of straight teeth spur gears is to transmit 20 kW when the pinion rotates at 300 rpm. The velocity ratio is 1:3. The allowable static stresses for the pinion and gear materials are 120 MPa and 100 MPa respectively. The pinion has 15 teeth and its face width is 14 times the module. Determine :

- i) Module
- ii) Face width and
- iii) Pitch circle diameter of both the pinion and the gear from the stand point of strength only taking into consideration the effect of the dynamic loading. Assuming steady load conditions and 8 – 10 hours of service per day, and the Service Factor (SF) as 1.

The tooth form factor Y can be taken as $Y = 0.154 - \left(\frac{0.912}{\text{No. of teeth}} \right)$

and the velocity factor C_v as $C_v = \frac{3}{3 + V}$ where V is expressed in m/s.

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- (b) Discuss the key properties of a smart material and explain the working of Shape Memory Alloys (SMAs) and their applications. **15**
- (c) Discuss the ABC analysis and EOQ model in inventory management and its importance. **10**

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- Q3.** (a) A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10000 N-m. The shaft is made of 45C8 steel having ultimate tensile stress of 700 MPa and ultimate shear stress of 500 MPa. Assuming a factor of safety as 6, determine the diameter of the shaft. **15**
- (b) Explain the key components of CIM system and role of Flexible Manufacturing Systems (FMS) in CIM. **15**
- (c) Explain different components of gating system with a neat sketch and the functions of runner and riser. **10**

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- Q4.** (a) There are 5 jobs namely A, B, C, D and E. These are to be assigned to 5 machines P, Q, R, S and T to minimize the cost of production. The cost matrix is given below. Assign the jobs to machine on one to one basis. **15**

Jobs (Cost in Rs.) →					
	A	B	C	D	E
Machines ↓					
P	8	7	4	11	6
Q	10	5	5	13	7
R	6	9	8	7	12
S	6	7	2	3	2
T	7	8	8	10	5

- (b) List the different 3D printing techniques and explain Stereolithography (SLA) process with neat sketch and its advantage. **15**
- (c) Discuss the principle of EDM with sketch, its process parameters and applications. **10**
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**SECTION – B****Q5.** Solve **any five** out of seven.**(8×5=40)**

- (a) Write assumptions made in Steady Flow Energy Equation (SFEE).
The mass flow rate in to air compressor is 810 kg/hr and the power input to the compressor is 55 kW.
The following data are known for the air entering and leaving conditions of the compressor :

	Inlet condition	Outlet condition
Pressure (bar)	1	10
Sp. volume (m^3/kg)	0.8	0.13
Velocity (m/s)	30	60
Elevation above the datum line (m)	2	3

The heat rejected to cooling water is 9.5 kW.

Determine :

- Increase in specific internal energy for the process.
 - Inlet and outlet area.
- (b) Write five criterion for site selection of nuclear power plant.
- (c) Explain pre-ignition detonation in S.I. engine.
- (d) Enumerate key components and their functions, working principle, advantage of CRDI technology.
- (e) Explain ODP and GWP. Discuss impact on refrigerator industry.
- (f) Derive an expression for efficiency of the insulated tip fin with usual parameters.
- (g) Explain terms of Navier-Stokes equation.

Q6. (a) 2000 kJ/min. of heat is supplied to a system at 500 K from a source at 1000 K. The temperature of the atmosphere is 27°C. Assuming the temperatures of system and source remains constant during heat transfer, find :

- The entropy production accompanying the heat transfer. **7**
- The decrease in available energy after heat transfer. **8**

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- (b) The following data is given for a 4 stroke, 4 cylinder diesel engine :

Diameter of cylinder = 35 cm, Piston stroke = 40 cm, Speed = 315 rpm, Indicated mean effective pressure = 7 bar, B.P. of engine = 260 kW, Fuel consumption = 80 kg/hr, C.V. of fuel = 43,000 kJ/kg, Hydrogen content in fuel = 13% and remaining is carbon, Air consumption = 30 kg/min, Cooling water circulated = 90 kg/min, Rise in temperature of cooling water = 38°C, Piston cooling oil used = 45 kg/min, Rise in temperature of cooling oil = 23°C, $C_{p_{oil}} = 2.2$ kJ/kg.K, Exhaust gas temp. = 322°C, $C_{p_{Exhaust}} = 1.1$ kJ/kg.K, Ambient temperature = 22°C, $C_{p_{Steam}} = 2$ kJ/kg.K, Latent heat of steam = 2520 kJ/K.

Find :

- i) The mechanical and indicated thermal efficiency.
 - ii) Draw up heat balance sheet on minute basis and percentage basis. **15**
- (c) Explain working of thermostatic expansion valve in VCRS. Why it is suitable for variable thermal load conditions ? **10**

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- Q7.** (a) Draw the general layout of thermal power plant using coal as fuel and explain working of coal and ash circuit, air and gas circuit, feed water and steam circuit and cooling water circuit. **15**
- (b) A hydro carbon fuel contains 86% carbon and 13% hydrogen by mass and the remainder is incombustible material. 25 kg of air is supplied per kg of fuel. Determine the dry and wet volumetric analysis of exhaust gases. Assuming the vapour formed to be a perfect gas and total pressure of the exhaust gas as 1.03 bar. Find the partial pressure of vapour in the exhaust gas and the heat carried by exhaust gas and steam per kg of fuel
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burned. If temperature of exhaust gas is 420°C and room temperature is 20°C . C_v for dry gas = 1 kJ/kg.K , C_v for steam = 2.1 kJ/kg.K .

Properties of steam at 0.0746 bar

$h_{\text{sat}} = 2576 \text{ kJ/kg}$, $T_s = 40^{\circ}\text{C}$. **15**

(c) Explain Fanno lines and Rayleigh lines. **10**

Q8. (a) Write ten desirable properties of refrigerants used in VCRS. Enlist 05 properties of refrigerant-absorbent pair used in vapour absorption system. **15**

(b) The air flow through axial compressor – The ambient conditions at inlet are 20°C and 1 bar . At exit, the total head temperature and pressure are 150°C and 3.5 bar and static pressure at exit is 3 bar .

Calculate :

- i) The isentropic efficiency.
- ii) The polytropic efficiency.
- iii) The air velocity at exit. **15**

(c) Air at a temperature of 290 K , flows in a centrifugal compressor running at 20000 rpm , slip factor = 0.8 , $\eta_{t-t} = 0.80$, $d_2 = 0.6 \text{ m}$. Assuming that the absolute velocities at the inlet and outlet are same.

Calculate :

- i) The temperature rise of air passing through the compressor.
 - ii) Stage pressure ratio. **10**
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